

TATA CONSULTANCY SERVICES

EXECUTIVE REPORT

TCS ACTIVE SOVEREIGNTY · TRANSPORT & MOBILITY

The Minimum Viable

Rail Infrastructure Manager

What must never stop — **the track is available and safe** — and how to keep it running under your own control.

01 EXECUTIVE SUMMARY

Sovereignty is now a board decision

Every organisation has a short list of things that must never stop. For a rail infrastructure manager, it starts with **the track is available and safe**. Active Sovereignty keeps that list running — under your own control — whatever happens to suppliers, jurisdictions or the threat landscape.

Sovereignty used to be a compliance question about where data is stored. It is now a strategic question of continuity and control: who could switch off, read or change the systems the business depends on — and how fast it could recover if they did. For transport operators, NIS2 and the CER Directive classify much of the sector as essential, and passenger expectations leave little tolerance for outages.

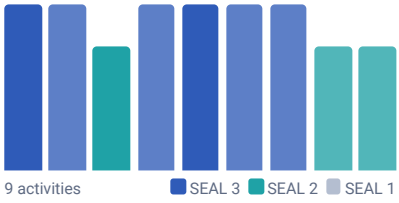
The answer is not to move everything into a sovereign environment. It is to define the **Minimum Viable Enterprise (MVE)**: the smallest set of capabilities, data and operations that must stay available and under European control. For a rail infrastructure manager, **6 of 9** activities form that load-bearing core, led by signalling & traffic control (ertms), capacity allocation & path planning and incident & disruption management — including the often-overlooked backbone of collaboration and IT operations.

One size does not fit all. **5 of 9** activities stay on public or public sovereign cloud; only the core moves to private or national. No activity is targeted above SEAL 3 — chips, hardware and much software and SaaS remain non-EU.

TCS recommends what is right for each activity; the board decides what to implement, what to phase and which risks to accept. The cost is driven by how the model is operated far more than by infrastructure — and it can be contained.

AT A GLANCE

9	value-chain activities, including the enterprise backbone
6	load-bearing — target band reaching SEAL 3
5	on public and public sovereign cloud
90	days to an executive decision that sets the direction



THREE MESSAGES FOR THE BOARD

1

Define what must never stop.

Agree the MVE and its recovery objectives — the commitments you can keep on your worst day.

2

Protect the core, not everything.

SEAL 3 only where criticality demands it; the rest on public cloud with proportionate controls.

3

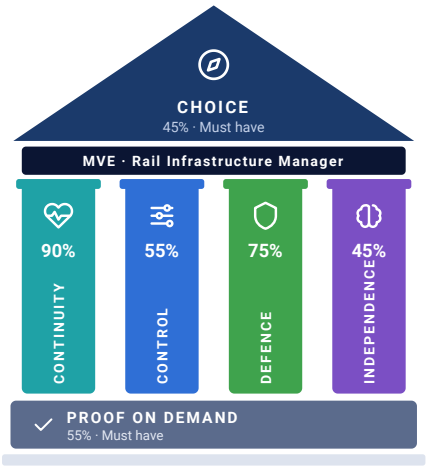
Make it provable.

Treat exit, recovery and control as capabilities you evidence continuously — not clauses in a contract.

02 The six questions
03 The path to your MVE
04 Risk posture
05 Resilience & operating model
06 The business case
07 What TCS brings
08 The roadmap
09 Design MVE · 90 days

02 THE MVE AND ITS SIX QUESTIONS

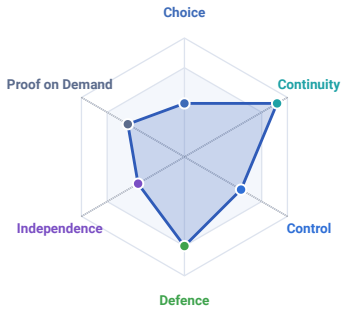
Six questions only the board can answer



At the centre sits the **Minimum Viable Enterprise** — for a rail infrastructure manager, **the track is available and safe** — held up by six pillars. Each is a question that sits with a different member of the executive team; no single function can answer it alone.

The percentages are the **TCS recommended focus for Transport & Mobility**: how much weight each question deserves on a scale from nice to have to critical. The board can adjust them; the minimum answers below cannot be left open.

FIGURE 1 The Enterprise House and the TCS recommended focus



Sovereignty fingerprint · Transport & Mobility

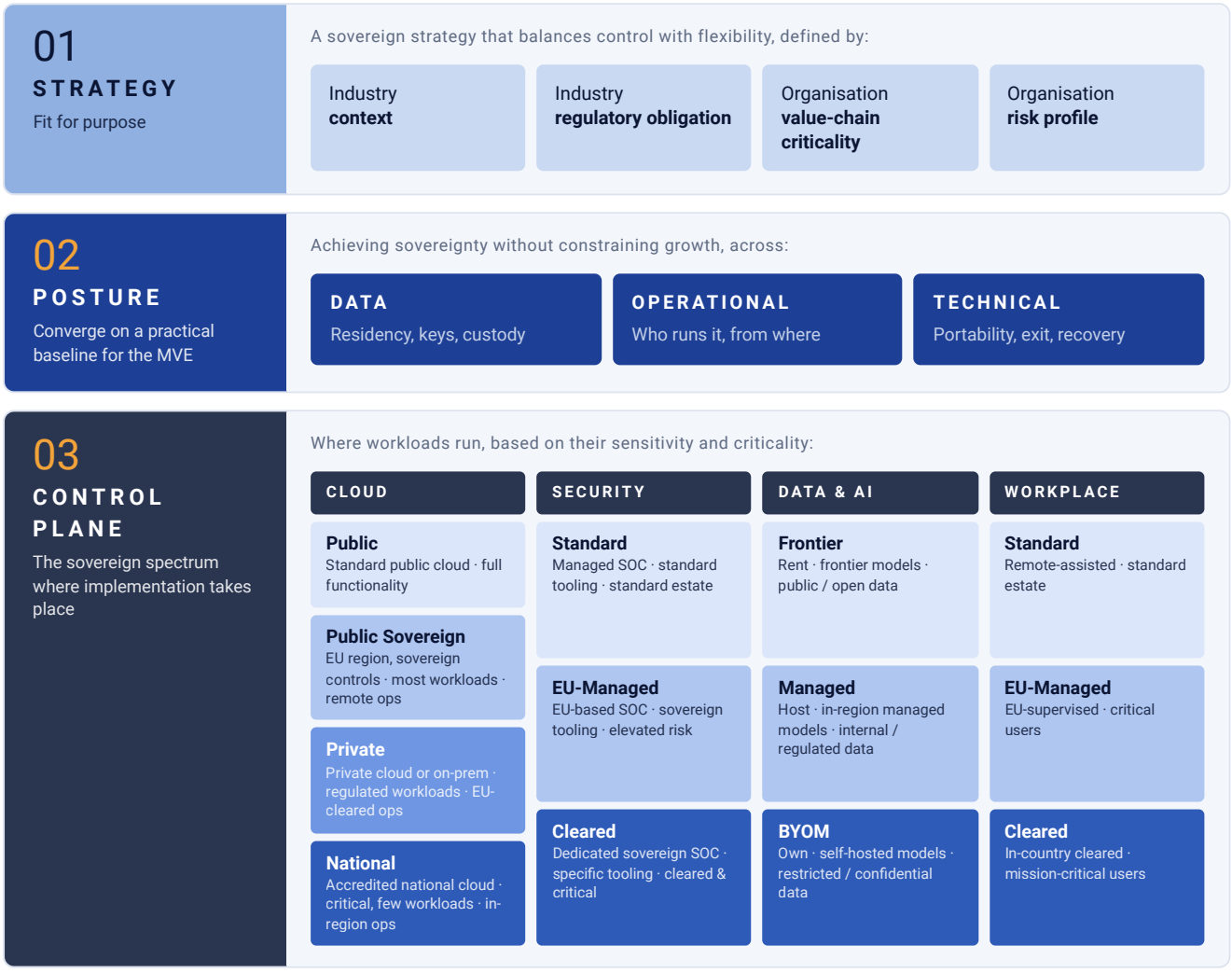
THE QUESTION	THE MINIMUM ANSWER FOR A RAIL INFRASTRUCTURE MANAGER	TCS FOCUS
Whoever changes the rules, can we still choose our own path? CHOICE · CEO / BOARD	Signalling and capacity systems under national jurisdiction only.	45% MUST HAVE
Will the business keep running through shocks we don't control? CONTINUITY · COO	The network stays safe and controllable through cyber attack or provider cut-off.	90% CRITICAL
Do we command the technology we run on — or does our provider? CONTROL · CIO / CTO	Signalling run by cleared, in-country operators; OT islanded.	55% MUST HAVE
If we're attacked or cut off, can we defend and restore ourselves? DEFENCE · CISO	Air-gap capable signalling; isolated recovery of traffic control.	75% CRITICAL
Can we harness the best data and AI without becoming hostage to it? INDEPENDENCE · CDO / CAIO	Network, capacity and asset data owned by the infrastructure manager.	45% MUST HAVE
Can we prove sovereignty actively, on demand — not once a year? PROOF ON DEMAND · CRO / CCO	Evidence for ministries and every operator on the network.	55% MUST HAVE

03 TCS ACTIVE SOVEREIGNTY FRAMEWORK

The path to your Minimum Viable Enterprise

Active Sovereignty is sovereignty that is not only designed, but **continuously operated, controlled and demonstrably assured**. The path runs through three layers — from strategic intent, to a practical posture, to the control plane where it is implemented.

FIGURE 2 The path to MVE for Active Sovereignty



From measured exposure to enforceable control

01 Strategy — fit for purpose

Sovereignty is a continuum, not a switch. The right level depends on the industry, its regulation, how critical each part of the value chain is and the organisation's risk profile. For transport operators, NIS2 and the CER Directive classify much of the sector as essential, and passenger expectations leave little tolerance for outages.

02 Posture — a practical baseline

Every activity gets a target across three dimensions: **Data** — where it resides and who holds the keys; **Operations** — who runs it and from where; **Technology** — how it is recovered, moved or exited.

03 Control plane — where it becomes real

Each workload is placed across four planes. Most run on public or public sovereign cloud with sovereign controls; the control-critical core on private cloud or on-prem; national cloud only for strict national-control workloads.

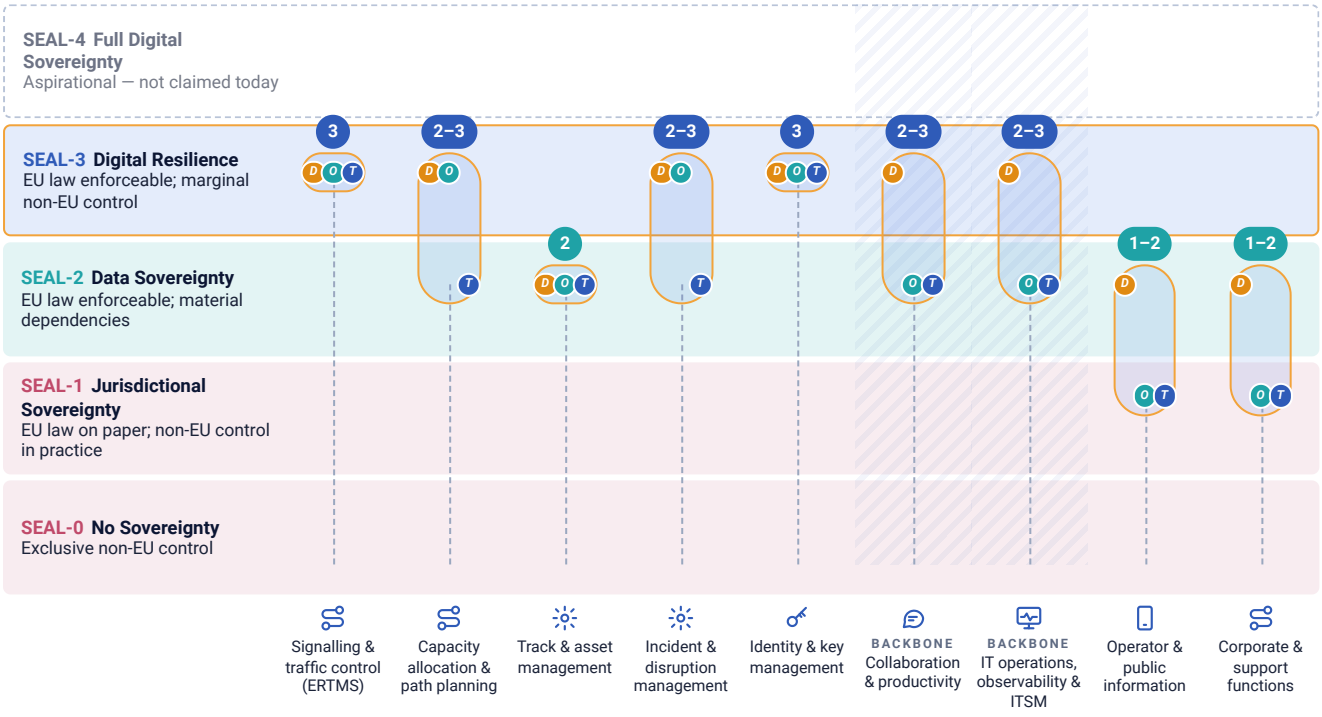
04 RISK POSTURE

Protection matched to criticality

SEAL-4 needs no critical non-EU dependency in chips, hardware, software or SaaS, and is not attainable today — so **no activity is targeted above SEAL 3**. Each activity below shows its Data, Operations and Technology targets as whole levels; the band spans them.

Recommendation based on the TCS Active Sovereignty Framework · Transport & Mobility

FIGURE 3 Target SEAL per activity — industry core plus the enterprise backbone (shaded)



ACTIVITY	TARGET	D · O · T	WHERE IT RUNS	WHY THIS TARGET
Signalling & traffic control (ERTMS)	SEAL 3	3 · 3 · 3	Private	Safety critical for every train.
Identity & key management	SEAL 3	3 · 3 · 3	Private	Root of trust.
Capacity allocation & path planning	SEAL 2–3	3 · 3 · 2	Private	Every operator depends on it.
Incident & disruption management	SEAL 2–3	3 · 3 · 2	Private	Restores service.
Collaboration & productivity ENTERPRISE BACKBONE	SEAL 2–3	3 · 2 · 2	Public Sovereign	Mail, files, chat and copilots (e.g. Microsoft 365, Teams, SharePoint, Copilot, Google Workspace) hold board papers…
IT operations, observability & ITSM ENTERPRISE BACKBONE	SEAL 2–3	3 · 2 · 2	Public Sovereign	Monitoring, logging, APM, CMDB and service-management tools hold the map of the estate: topology, credentials in…
Track & asset management	SEAL 2	2 · 2 · 2	Public Sovereign	Availability of the network.
Operator & public information	SEAL 1–2	2 · 1 · 1	Public	Access channel.
Corporate & support functions	SEAL 1–2	2 · 1 · 1	Public	Important, not load-bearing.

05 RESILIENCE & OPERATING MODEL

What holds under stress — and where it runs

A posture is only as good as its behaviour under stress. The MVE keeps the core running through each disruption below; where it adds real value, TCS recommends a **fallback on top**. We recommend — the board decides.

FIGURE 4 The rail infrastructure manager MVE as a deliberate mix — number of activities per tier, darker = more of the MVE



FIGURE 5 Disruption scenarios — what the recommended MVE delivers, and where a fallback is recommended

DISRUPTION	WITH THE RECOMMENDED MVE	RECOMMENDED FALLBACK
Cloud provider cut off *Without an MVE: Core services stop until the provider relationship is restored.*	The core already runs on EU-operated private and national environments; channels on public sovereign cloud fail over within agreed timeframes.	Continuity pod — a minimal standby of the core on a second EU-operated environment.
Ransomware hits the core *Without an MVE: Core systems and backups are encrypted; recovery time is unknown.*	An immutable vault and clean-room rebuild restore the core in agreed priority order.	Break-glass identity held outside the primary platform, so access can be restored first.
Foreign access to data or keys *Without an MVE: Data, documents, operational metadata and keys can be compelled outside EU law.*	Hold-your-own keys and EU-cleared operations remove the access path.	No fallback needed — the MVE holds.
AI model withdrawn *Without an MVE: AI-dependent decisions and copilots stop or degrade without warning.*	Critical AI workloads run on managed or owned models with portable data and prompts.	A privately deployed model (BYOM) as a switch-over target for the most critical AI use cases.
Critical SaaS provider exits *Without an MVE: Core processes, documents and their data are locked in an unavailable platform.*	Data is held in portable formats under your control, with a tested exit plan for each critical SaaS provider.	An in-country standby instance with daily data export for the processes that cannot stop.
Regulator asks for proof — today *Without an MVE: Evidence is assembled manually, under deadline pressure.*	Continuous evidence on control, exit and recovery is available on demand.	No fallback needed — the MVE holds.

06 THE BUSINESS CASE

Contained cost, not blanket cost

Boards rightly ask what sovereignty costs. The cost is driven far less by infrastructure than by the **operating model around it** — and most of it recurs every year. The decisive question is how to contain the recurring cost of running the MVE.

FIGURE 6 What drives the cost of a sovereign operating model — and how TCS contains it

What drives the cost up

Running & protecting the estate Local, security-cleared teams around the clock	*Ongoing — the largest line*
Identity & access control A separate, customer-controlled sign-in and key system	*Set-up + ongoing*
Security & isolation Dedicated protection for the most sensitive systems	*Set-up + ongoing*
Business continuity & recovery Independent, regularly tested recovery	*Set-up + ongoing*
Audit & compliance Evidence and independent checks	*Ongoing — grows as rules tighten*
Keeping data & keys in-region In-region data, backups and replication	*Set-up + ongoing*
Platform & infrastructure Dedicated capacity or more controlled ground	*Set-up + ongoing*
Keeping sovereignty current Re-certification and rehearsals	*Ongoing*
Software & suppliers Sovereign editions and vendor assurance	*Ongoing — only where needed*

How TCS contains it

TCS SOVEREIGNSECURE CLOUD™

- **Build-Public, Run-Sovereign** — build on flexible cloud; run hardened workloads sovereign
- **Shared Sovereign Platform** — cleared teams and infrastructure shared across organisations
- **Open Technology** — open-source core; sovereign editions only where required

SOVEREIGN OPERATIONS · TCS EDEN™

- **Autonomous Operations** — AI-led operations; cleared teams handle exceptions
- **AI Service Experience** — AI-first service desk
- **Always-On Resilience** — recovery continuously and automatically proven

SOVEREIGN SECURITY

- **Autonomous Security** — threats detected and contained automatically
- **Unified Sovereign Identity** — one identity and key platform

AUDIT & COMPLIANCE

- **Policy-as-Code** — evidence generated by the platform
- **Continuous Assurance** — automated drift detection

FIGURE 7 What a credible sovereignty business case shows

1 The cost of not acting
Outage, exit and supervisory exposure if the core fails or is cut off.

2 The overlay — for the MVE only
Sovereign controls on the load-bearing core, not the whole estate.

3 The value created
Continuity, market access, supervisor confidence and freedom of choice.

4 The funding gates
Investment released in waves, as risk and value demand.

WE CONTAIN THE COST; WE DO NOT CLAIM TO REMOVE IT

The first lever is the MVE itself: **protect the core, not everything**. Over five years the recurring cost of sovereign operations rises with the cost of living; as AI and automation mature, efficiency compounds and bends it down.

07 WHAT TCS BRINGS

Sovereignty where it matters, flexibility where it helps

Advisory, four implementation planes and sovereign operations under one governed model — **from executive intent to a running Active Sovereign MVE**, on a strong European operations backbone.

FIGURE 8 The TCS Active Sovereignty value proposition

Sovereignty Advisory

Assessment, MVE definition, target posture, operating model and business case; then continuous assurance. Across your sovereign landscape: multi-cloud, private, security, AI and workplace.

TCS SovereignSecure Cloud™

- National, private and public sovereign clouds
- TCS private cloud
- One orchestration, policy and FinOps layer

SOVEREIGN CLOUD ECOSYSTEM

- National sovereign cloud
- Private · TCS Enterprise Cloud
- Public sovereign · Azure, AWS, Google Cloud

Sovereign Security

- EU-managed and cleared security operations
- Hold-your-own keys and sovereign identity
- Isolation, air-gapping and cyber recovery

HIGHLIGHT

- Sovereign SOC — dedicated SOC for the EU, in Spain

Sovereign AI

- Frontier, managed or BYOM models
- Sovereign AI / LLM stack and LLMOps
- Data and model independence

EUROPEAN MODEL ECOSYSTEM

- Mistral AI
- Silo AI
- EuroLLM
- Aleph Alpha

Sovereign Workplace

- EU-managed, country-cleared, remote-assisted support
- AI-first service experience
- Workplace continuity under disruption

WORKPLACE OPTIONS

- M365 Local
- Continuity Pod — TCS Private Cloud
- Nextcloud
- Proton

TCS EDEN™ · THE OPERATIONAL BACKBONE

Sovereignty is only as strong as the operations behind it. TCS EDeN™ runs sovereign estates inside EU jurisdiction, by EU-based professionals — cloud-agnostic, secured and provable.

31+
LOCATIONS
across Europe

15,000+
PROFESSIONALS
EU-based, within EU jurisdiction

10
AVAILABILITY ZONES
TCS Private Cloud — plus operations on the public clouds

1
DEDICATED SOC
Sovereign SOC for the EU, in Spain

Advisory sets the direction. Every engagement starts with intent: which activities must hold, to what level, at what cost. Sovereignty Advisory turns that into the MVE, the target posture, the operating model and the business case — and later into continuous assurance, so the posture does not drift.

Four planes deliver it. TCS SovereignSecure Cloud™ — including TCS Private Cloud alongside national and public sovereign clouds — Sovereign Security with a dedicated SOC for the EU, Sovereign AI and Sovereign Workplace implement the posture, activity by activity, on the European ecosystem.

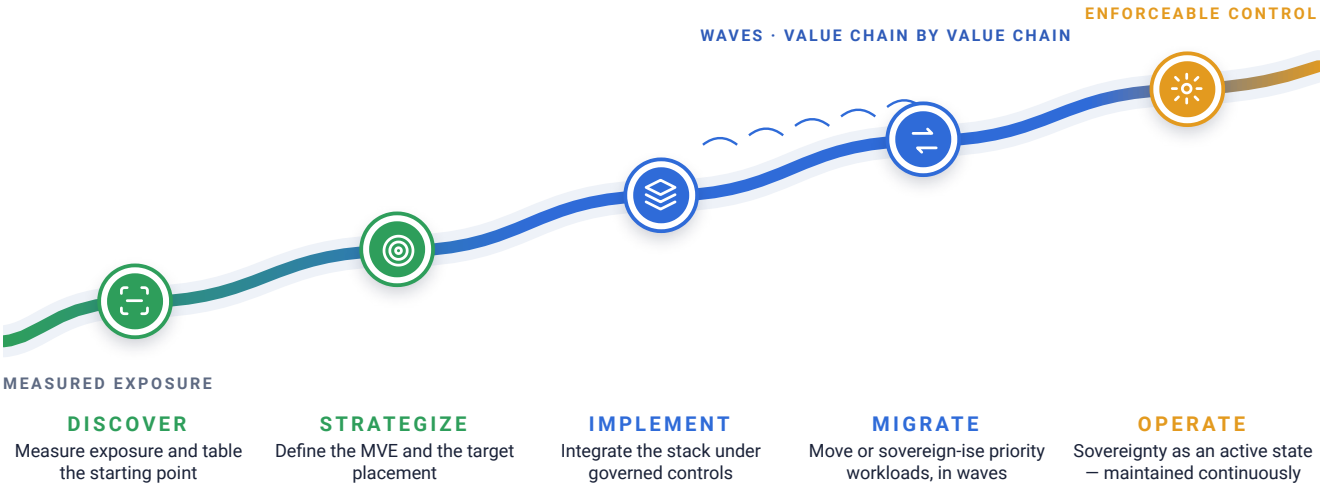
TCS EDeN™ runs it. Sovereignty is only as strong as the operations behind it: 31+ locations and 15,000+ EU-based professionals, 10 availability zones on TCS Private Cloud plus operations on the public clouds, and a dedicated SOC in Spain.

08 RUN ACTIVE SOVEREIGNTY WITH TCS

The roadmap to Active Sovereignty

A journey, not a project: design the MVE for a rail infrastructure manager, build resilience in waves — value chain by value chain — then operate sovereignty as **an active state, maintained continuously**, with proof on demand.

FIGURE 9 From measured exposure to enforceable control



Design MVE

Discover · Strategize · Months 0–3

OUTCOMES

- Starting point tabled: exposure measured
- MVE defined — enterprise-wide or one value chain
- Workload placement across the four tiers
- Executive decision: direction set

TCS ACCELERATORS

- TCS Active Sovereignty Framework
- Industry baselines & risk postures
- Sovereign Assessment Agents
- Sovereign Operating Model
- Sovereign Use Case Catalogue
- TCS Hybrid Cloud Advisor

Build Resilience

Implement · Migrate · ≈ Months 3–18 · in waves

Workloads · wave by wave
Keys & identity
Sovereign landing zone

OUTCOMES

- Sovereign landing zones in place
- Keys & identity under EU control
- Priority workloads migrated or sovereign-ised in place
- Exit runbooks tested

TCS ACCELERATORS

- TCS Sovereign Forge
- Agentic AI Infrastructure for Local
- TCS AgentVMove
- TCS SovereignGuard
- FinOps for AI
- TCS Agentic SDLC
- Agentic Migration Factory · MS · AWS · GCP

Operate Active Sovereignty

Operate · an active state · Continuous

OUTCOMES

- MVE implemented and active
- Posture maintained continuously — proof on demand
- EU SOC monitoring & resilience drills
- Exit readiness verified, drift corrected

TCS ACCELERATORS

- TCS Alpha
- TCS Iris
- TCS Xponence
- TCS Cognix
- TCS Agentic Operations
- MS Sovereign data controls
- AWS Sovereign Cloud Command Center
- GCP Suite

Indicative timeline — the pace is set per value chain in the assessment. Design MVE — the first stage — is where the starting point is tabled. The next page shows its two scopes.

09 DESIGN MVE · THE 90-DAY ACCELERATION

Two ways in, one outcome: the direction set

Discover and Strategize, powered by the TCS Sovereignty Assessment, turn the reference MVE in this report into your own — scored on your evidence. It runs in one of two scopes; **the deliverables and the executive decision pack change with it**.

Before day one
Agentic pre-assessment — readiness, data, regulation, risk-posture snapshot

Discovery sprint
Measure exposure and table the starting point

Strategy sprint
Define the MVE, the placement and the operating model

Day 90
Direction set

SCOPE · ENTERPRISE-WIDE

Board sovereignty strategy

Across the whole enterprise: place every workload on the right tier and give the board one sovereignty strategy. Execution then runs in waves, value chain by value chain.

DISCOVERY SPRINT

Measure enterprise exposure

- Enterprise workload inventory & criticality
- Exposure heatmap across all value chains
- Data residency & key control, estate-wide
- Regulatory & extraterritorial exposure
- Provider concentration & vendor evidence

STRATEGY SPRINT

Define the enterprise MVE

- Workload placement across the four tiers
- Enterprise MVE & D·O·T target bands
- Sovereign operating model & guardrails
- Identity, keys, exit & portability principles
- Roadmap waves & investment envelope

Design sprint · Optional · in parallel — Design 2–3 pilots across the planes, so the strategy is grounded in your own estate.

EXECUTIVE DECISION PACK · ENTERPRISE-WIDE

Board sovereignty strategy

- **Sovereignty strategy & risk appetite** — What the board approves, and why now.
- **Workload placement principles** — Which workloads go to which tier.
- **Exposure today vs target** — Enterprise heatmap, current versus target.
- **Investment envelope** — Cost, trade-offs and funding gates.
- **Wave plan** — Which value chains first, what follows.

Fits when the board needs one sovereignty strategy for the whole enterprise before execution starts.

SCOPE · ONE VALUE CHAIN

Executive proof point

On one value chain: go deep end to end and give the executives a proof point — then scale the approach to the whole enterprise.

DISCOVERY SPRINT

Map one value chain end to end

- Activities, data flows & dependencies
- Regulatory exposure of the value chain
- Vendor & SaaS evidence for its providers
- Disruption impact, activity by activity
- Baseline D·O·T per activity

STRATEGY SPRINT

Design the value-chain MVE

- Target D·O·T band per activity
- Deployment design across the tiers
- Continuity & exit design
- Identity & key design for the chain
- Business case for the value chain

Proof-point design · Core · ready to build — Design the proof point in detail — architecture, controls and a test plan — ready to build in the next phase.

EXECUTIVE DECISION PACK · ONE VALUE CHAIN

Executive proof point

- **Go / no-go on the proof point** — What to build, and why this value chain.
- **Value-chain MVE design** — Target bands and deployment per activity.
- **Value-chain business case** — Cost, benefit and risk accepted.
- **Exposure today vs target** — For this value chain.
- **How to scale** — From one value chain to the enterprise.

Fits when executives need a visible proof point on one critical value chain before scaling.

NEXT STEP

Start with a conversation about your MVE.

Talk to your TCS team — and bring an executive decision on your Minimum Viable Enterprise to the table in 90 days.

TATA CONSULTANCY SERVICES